

NBCA - 201
DATA STRUCTURE

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COURSE OUTCOMES (COs)

After the completion of the course, students are expected to have the ability to:

1. Learn how to represent arrays, linked lists, stacks, queues in memory using the algorithms and their common applications.
2. Understanding the concept of recursion, application of recursion and its implementation and removal of recursion.
3. Learn the computational efficiency of the sorting and searching algorithms.
4. Learn implementation of Trees and Graphs, and various operations on these data structure.
5. Identify the alternative implementations of data structures with respect to its performance to solve a real world problem.

Unit-I

07

Introduction: Basic Terminology, Elementary Data Organization, Built in Data Types, Abstract Data Types. **Arrays:** Derivation of Index Formulae for 1D, 2D, 3D & nD Array, Application of arrays. **Linked lists:** Implementation of Singly Linked List and Doubly Linked List.

Unit-II

08

Stacks: Basic operations: Push & Pop, Array implementation of Stack in C, Application of stack: Prefix and Postfix Expressions, and Evaluation of postfix expression. **Queues:** Basic operations: Create, Add, Delete, Circular queues, and Array implementation of queues in C.

Unit-III

09

Trees: Basic terminology, Binary Trees, Binary Tree Representation: Array and Pointer (Linked List) Representation, Binary Search Tree, Strictly Binary Tree, and Complete Binary Tree. Extended Binary Trees, Tree Traversal algorithms: In-order, Pre-order and Post-order, Constructing Binary Tree from given Tree Traversal and Insertion.

Unit-IV

08

Searching: Sequential search and Binary Search. **Sorting:** Insertion Sort, Selection, Bubble Sort, Quick Sort, and Radix Sort.

Unit-V

08

Graphs: Basic terminology, Graph Representations: Adjacency Matrices, Adjacency List. Graph Traversal: Depth First Search and Breadth First Search, Connected Component, Spanning Trees, Minimum Cost Spanning Trees: Prims and Kruskal algorithm.

Text Books:

1. Aaron M. Tenenbaum, Yedidiah Langsam and Moshe J. Augenstein “Data Structures Using C and C++”, PHI

2. R. Kruse et al, "Data Structures and Program Design in C", Pearson Education
3. Thareja, "Data Structure Using C" Oxford Higher Education

Reference Books:

1. Horowitz and Sahani, "Fundamentals of Data Structures", Galgotia Publication
2. R. Kruse et al, "Data Structures and Program Design in C", Pearson Education
3. Lipschutz, "Data Structures" Schaum"s Outline Series, TMH

NBCA-206P
DATA STRUCTURE LAB

L T P

0 0 3

1. To implement addition and multiplication of two 2D arrays.
2. To transpose a 2D array.
3. To implement stack using array.
4. To implement stack using linked list.
5. To implement queue using array.
6. To implement queue using linked list.
7. To implement circular queue using array.
8. To implement circular queue using linked list.
9. To implement binary tree using linked list.
10. To implement binary search tree using linked list.
11. To implement tree traversals using linked list.
12. To implement BFS using linked list.
13. To implement DFS using linked list.
14. To implement Linear Search.
15. To implement Binary Search.
16. To implement Bubble Sorting.
17. To implement Selection Sorting.
18. To implement Insertion Sorting.

Note: The Instructor may add/delete/modify/tune experiments, wherever he/she feels in a justified manner.